Numerical Methods

ANSWER KEY

Section 1: Location of Roots & Sign Change

1 Show that the equation $x^3 - 2x - 5 = 0$ has a real root in the interval $[2, 3]$.

a. $f(2) = -1 < 0$ **and** $f(3) = 16 > 0$**, so by IVT a root exists** ✓

b. $f(2) = 1 > 0$ and $f(3) = -2 < 0$

c. $f'(x) > 0$ for all x

d. $f(2.5) = 0$

$f(2) = 8 - 4 - 5 = -1 < 0$. $f(3) = 27 - 6 - 5 = 16 > 0$. Since f is continuous on $[2, 3]$ and changes sign, the Intermediate Value Theorem guarantees at least one root.

2 Under which condition does a sign change of $f(x)$ between a and b fail to guarantee a root in (a, b)?

a. **When** $f(x)$ **is not continuous on** $[a, b]$ ✓

b. When $f'(x)$ changes sign

c. When $f(x)$ has an inflection point

d. When the interval is wider than 1 unit

If $f(x)$ has a vertical asymptote or jump discontinuity in (a, b) (e.g. $f(x) = 1/x$ between -1 and 1), a sign change occurs without any root.

3 Use the sign change method to determine which interval of length 0.1 contains the root of $x^3 = 20$.

a. $[2.7, 2.8]$ ✓

b. $[2.5, 2.6]$

c. $[2.6, 2.7]$

d. $[2.8, 2.9]$

$2.7^3 = 19.683 < 20$. $2.8^3 = 21.952 > 20$. The root lies in $[2.7, 2.8]$.

Section 2: Fixed-Point Iteration $x_{n+1} = g(x_n)$

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4 Rearrange the equation $x^3 - x - 1 = 0$ into an iterative formula $x_{n+1} = g(x_n)$.

a. $x_{n+1} = \sqrt[3]{x_n + 1}$ ✓

b. $x_{n+1} = x_n^3 - 1$

c. $x_{n+1} = \dfrac{1}{x_n^2 - 1}$

d. $x_{n+1} = x_n^3 + x_n$

$x^3 = x + 1 \Rightarrow x = (x+1)^{1/3}$. Iterative formula is $x_{n+1} = \sqrt[3]{x_n + 1}$.

5 State the condition on $g'(x)$ for the iteration $x_{n+1} = g(x_n)$ to converge to a root α.

a. $|g'(\alpha)| < 1$ ✓

b. $|g'(\alpha)| > 1$

c. $g'(\alpha) = 0$

d. $g'(\alpha) > 0$

Fixed-point iteration converges if and only if $|g'(x)| < 1$ in a neighborhood of the root α.

6 Starting with $x_1 = 1.3$, use $x_{n+1} = \sqrt[3]{x_n + 1}$ to calculate x_2 and x_3 correct to 3 decimal places.

a. $x_2 = 1.320, x_3 = 1.324$ ✓

b. $x_2 = 1.300, x_3 = 1.310$

c. $x_2 = 1.350, x_3 = 1.380$

d. $x_2 = 1.315, x_3 = 1.330$

$x_2 = (2.3)^{1/3} \approx 1.3200$. $x_3 = (2.3200)^{1/3} \approx 1.3238 \approx 1.324$.

Section 3: Cobweb and Staircase Diagrams

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7 If $0 < g'(\alpha) < 1$ at the root, what pattern is produced on the iteration diagram?

a. **A staircase diagram converging to the root ✓**

b. A cobweb diagram converging to the root

c. A diverging staircase diagram

d. An oscillating cycle

When the gradient $g'(x)$ is positive and less than 1, successive iterations approach the root monotonically from one side, forming a staircase diagram.

8 If $-1 < g'(\alpha) < 0$ at the root, what pattern is produced on the iteration diagram?

a. **A cobweb diagram spiraling inwards (converging) ✓**

b. A staircase diagram spiraling outwards

c. A cobweb diagram spiraling outwards (diverging)

d. A vertical asymptote

A negative derivative between -1 and 0 causes the iterates to alternate sides around the root while decreasing in distance, forming an inward-spiraling cobweb.

9 What happens when $|g'(\alpha)| > 1$?

a. **The iteration diverges away from the root ✓**

b. The iteration converges very quickly

c. The iteration oscillates between two fixed values

d. The root becomes imaginary

If $|g'(\alpha)| > 1$, the fixed point is repulsive (unstable), and the sequence diverges away from α.

Section 4: Newton-Raphson & Advanced Applications

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10 State the Newton-Raphson iteration formula for finding roots of $f(x) = 0$.

a. $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$ ✓

b. $x_{n+1} = x_n + \dfrac{f(x_n)}{f'(x_n)}$

c. $x_{n+1} = x_n - \dfrac{f'(x_n)}{f(x_n)}$

d. $x_{n+1} = \dfrac{x_n f'(x_n) - f(x_n)}{2}$

The Newton-Raphson formula is $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$.

11 Use one iteration of Newton-Raphson with $x_0 = 2$ to approximate the root of $f(x) = x^2 - 5$.

a. $x_1 = 2.25$ ✓

b. $x_1 = 2.20$

c. $x_1 = 2.30$

d. $x_1 = 2.15$

$f(2) = 4 - 5 = -1$. $f'(x) = 2x \Rightarrow f'(2) = 4$. $x_1 = 2 - \frac{-1}{4} = 2 + 0.25 = 2.25$.

12 Under which circumstance will the Newton-Raphson method fail completely at an iteration step x_n?

a. **When $f'(x_n) = 0$ (tangent is horizontal)** ✓

b. When $f(x_n) = 0$

c. When $f''(x_n) = 0$

d. When x_n is negative

If $f'(x_n) = 0$, the tangent line is horizontal and never intersects the x-axis, causing division by zero in the formula.

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